

MUSAWANDILE SHAMASE

Junior Electronic Design & Detailing Engineer Candidate | BEngTech Electronics & Computer Engineering

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PROFESSIONAL SUMMARY

Electronics and Computer Engineering (BEngTech, DUT) graduate with a strong technical foundation in electronic circuit design, embedded systems, and control systems, gained through coursework and hands-on prototyping with microcontrollers, sensors, and wiring/harness-style builds. Comfortable working from engineering requirements and drawings through to a tested, working build, with attention to component selection, reliability, and electrical safety. Currently completing a PLC Programming Certification to strengthen industrial automation knowledge. Seeking a Junior Electronic Design & Detailing Engineer role to apply this foundation to electrical/electronic harness design, prototyping, and production, friendly documentation.

TECHNICAL SKILLS

- Electronics & Hardware: Electronic circuit design & troubleshooting, digital & analogue electronics, embedded systems, PCB assembly (SMT & through-hole), soldering, wiring/harness prototyping
- Engineering Design Considerations: Component sourcing & selection, reliability, electrical safety awareness, routing and assembly of wired sub-systems (from project prototyping work)
- Industrial Automation: PLC architecture, scan cycles, digital & analog I/O configuration, ladder logic, timers & counters
- Control Systems & Instrumentation: Control systems theory, process instrumentation, signals & systems, power engineering fundamentals
- Programming: C/C++, Arduino C++, Python, Ladder Logic (in progress)
- Software & Tools: MATLAB, Arduino IDE, Microsoft Office; open to developing Creo Parametric / 3D CAD proficiency
- Instruments & Testing: Oscilloscope, digital multimeter, calibration & test equipment

RELEVANT ACADEMIC COURSEWORK (BENGTECH, DUT)

- Electronic Circuit Design 2A & 2B — Pass Distinction / Pass
- Analogue & Digital Electronics IA/IB — Pass / Pass Distinction
- Fundamentals of Instrumentation 2A; Process Instrumentation 3A
- Fundamentals of Control Systems 2B (Pass Distinction); Control Systems 3A; Process Control Systems 3B
- Fundamentals of Power Engineering 2A; Renewable Energy 3B
- Embedded Systems 2B; Fundamentals of Microcontrollers 2A (Pass Distinction)
- Electronic Design Project 3A & 3B; Engineering Ethics and Professional Skills 3B (Pass Distinction)

ENGINEERING PROJECTS

Smart Home Irrigation System

Arduino, C++, Soil Sensors, Relay Modules

- Designed and prototyped an automated irrigation control system that monitors soil moisture in real time to regulate water pump actuation, including wiring of sensors, relays, and pump circuitry.
- Programmed microcontrollers in C++ to optimise irrigation intervals, reducing water waste and improving system efficiency.

Cable Fault Detection System

Embedded Systems, Signal Processing, Sensors

- Engineered a diagnostic system to detect and pinpoint fault locations in electrical and signal cabling, reducing operational downtime — directly relevant to harness reliability and electrical safety.
- Integrated embedded sensor interfaces and signal processing logic to deliver precise fault detection and reporting.

Car Safety Interlock System

Microcontrollers, Embedded C, Safety Logic

- Developed an embedded safety interlock system that inhibits vehicle startup until safety checks (e.g. seatbelt fastening) are validated, with focus on wiring integrity and electrical safety compliance.
- Configured microcontroller hardware interrupts and firmware logic in Embedded C for real-time safety condition verification.

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

- PLC Programming Certification — PLCDOJO (In Progress): PLC architecture, scan cycles, digital/analog I/O, ladder logic, timers & counters, industrial automation.

WORK EXPERIENCE

Merchandiser

April 2026 – Present

MyChina, Boksburg, South Africa

- Maintain product displays, stock replenishment, and inventory accuracy in a high-volume retail environment.
- Demonstrate operational discipline, attention to detail, and effective team coordination to consistently achieve daily targets.

Educational Assistant

June 2025 – November 2025

KwaBulawayo High School (Department of Basic Education), South Africa

- Supported educators with lesson preparation, classroom administration, and academic tracking.
- Delivered structured academic guidance and tutoring to high school learners in core STEM subjects.

Volunteer Programming Tutor

March 2022 – August 2024

Dataholics Student Organisation, Durban, South Africa

- Tutored undergraduate students in computer programming fundamentals, logic design, and software debugging.
- Conducted practical coding workshops and test preparation sessions that improved academic outcomes.

EDUCATION

Bachelor of Engineering Technology (BEngTech) — Electronics & Computer Engineering

Durban University of Technology (DUT), Durban, South Africa — Graduated 2025

ADDITIONAL INFORMATION

- Driver's Licence: Code 10 (C1)
- Relocation: Open and available to relocate across South Africa
- Work Eligibility: South African Citizen — eligible to work immediately
- References available on request